Product Requirements Document (PRD): Tara Fitness Center

1. **Project Overview**

****Tara Fitness Center**** is a modern, full-stack (MERN) web application designed to digitize gym operations, improve user engagement, and streamline member management. It provides a premium digital experience for both the gym owner (Admin) and its members (Users).

2. **Target Audience**

- ****Members (Users)****: Accessing membership plans, tracking their status, generating AI nutrition plans, and managing profiles.

- ****Gym Owner (Admin)****: Managing users, verifying payments, updating content (gallery, trainers), and viewing business analytics.

3. **Core Features**

🔐 **Authentication & Security**

- ****Email/Password Authentication****: Secure login and registration with hashed passwords (bcrypt).

- ****Google OAuth****: Fast login via Google.

- ****Password Recovery****: Secure 'Forgot Password' flow with expiring reset tokens sent via email.

- ****JWT Protection****: All private routes and API endpoints are protected using JSON Web Tokens.

- ****Role-Based Access Control (RBAC)****: Distinct permissions for `Admin` and `User` roles.

👤 **Member Dashboard (User Portal)**

- ****Real-Time Status****: View current membership status (Active, Pending, Expired) with countdown timers.

- ****Profile Management****: Update name, contact details, date of birth, and training goals.

- ****Cloudinary Image Upload****: High-performance profile photo uploads with automatic compression and Cloudinary storage.

- **AI SmartNutrition (Groq AI)**: Generate personalized daily calorie and macro targets based on age, weight, height, activity level, and goals.

💳 **Membership & Payments**

- **Flexible Plans**: Separate categories for Basic Training, Personal Training, and Steam Bath packages.

- **Secure Payment Flow**: Support for UPI (QR code) and Cash-based membership purchases.

- **OCR-based Fraud Protection**: Backend verification system that checks transaction details (image/text) to prevent duplicate or falsified submissions.

- **Admin Verification**: All payments require final administrative approval before memberships are activated.

🛡️ **Admin Command Center**

- **Live Analytics**: Real-time stats on total members, active memberships, and recent revenue.

- **User Management**: Search, edit, and manually create/delete user accounts.

- **Content Management System (CMS)**:

  - **Live Gallery**: Upload and manage gym photos.

  - **Trainer Profiles**: Add/Edit trainers with certifications and Instagram links.

  - **Offer Management**: Post and update the latest gym discounts.

- **Real-Time Notifications**: Integrated with Socket.io for instant database change updates across all admin panels.

🏢 **Public Branding & Engagement**

- **Responsive Landing Page**: Sleek, dark-themed design using Framer Motion animations.

- **Interactive Trainer Gallery**: Showcase professional staff.

- **Package Browsing**: Explore different gym plans before signing up.

- **Dynamic Reviews**: High-rated feedback automatically pulled into the testimonial section.

- **Enquiry System**: Integrated WhatsApp-ready contact forms.

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4. **Technical Stack**

Frontend

- **React (Vite)**: Core framework for a fast, modern UI.

- **Vanilla CSS / Inline Styles**: Custom, premium dark-theme design.

- **Framer Motion**: Smooth entry animations and micro-interactions.

- **Lucide React**: Vector icons and visual indicators.

- **React Router Dom**: Dynamic client-side routing.

Backend

- **Node.js & Express**: High-performance RESTful API.

- **Socket.io**: Real-time bi-directional events for dashboard synchronization.

- **Mongoose (MongoDB)**: Scalable, document-oriented database.

- **Cloudinary SDK**: Modern cloud-based image handling.

- **Nodemailer**: Transactional and recovery emails.

Security Layers

- **Helmet**: 15+ secure HTTP headers configured.

- **Rate Limiter**: Brute-force protection for Auth routes and AI generation limiters.

- **CORS Whitelisting**: Strict origin controls (only allowing tfc365.com).

- **Express-Mongo-Sanitize**: Prevention of NoSQL injection attacks.

- **Global Error Handling**: Production-safe error concealment (NODE_ENV=production).

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5. **Internal Infrastructure**

- **Environment Management**: Strict `.env` configuration for sensitive keys (MongoDB URI, JWT Secret, Groq API Key).

- **Deployment Strategy**:

  - **Frontend**: Vercel (Production) with custom domain `tfc365.com`.

  - **Backend**: Render (Production) for server-side logic and database persistence.

- **Automated Workflows**: Cron jobs for membership expiration checks and reminders.

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6. **Future Roadmap**

- **Mobile App**: PWA conversion or Flutter/React Native application.
- **WhatsApp API Integration**: Automated enrollment and payment receipt messages.
- **Attendance Tracking**: QR code scanning system at the gym entrance.
- **E-commerce**: Integrated store for supplements and equipment.

Status: V1.0.0 (Production Live)

Last Updated: April 3, 2026